Movie Recommendation System Project Report

1. Abstract

This project presents a Movie Recommendation System developed using Machine Learning techniques. The system recommends movies to users based on their preferences such as selected movie, genre, and ratings. It uses a content-based filtering approach along with rating-based filtering to provide accurate and relevant suggestions. A Streamlit-based web interface is implemented to make the system interactive and user-friendly.

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2. Introduction

With the rapid growth of online streaming platforms, users often face difficulty in selecting movies due to the vast amount of available content. Recommendation systems help solve this problem by suggesting relevant items based on user preferences.

This project focuses on building a movie recommendation system using fundamental AI/ML concepts learned during the course. The system is simple, efficient, and demonstrates real-world application of machine learning.

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3. Problem Statement

Users struggle to find movies that match their interests due to the large number of available options. The goal of this project is to design a system that can recommend movies based on:

- Similarity to a selected movie

- Preferred genre/category
- Minimum rating threshold

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4. Objectives

- To build a recommendation system using machine learning
- To apply content-based filtering techniques
- To provide genre-based and rating-based filtering
- To develop an interactive user interface using Streamlit
- To enhance user experience in movie selection

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5. Methodology

5.1 Data Collection

A movie dataset containing features like movie title, genres, and ratings is used.

5.2 Data Preprocessing

- Handling missing values
- Converting text data into numerical form
- Combining relevant features

5.3 Feature Extraction

TF-IDF vectorization is used to convert textual data (genres) into numerical vectors.

5.4 Model Building

Cosine similarity is used to compute similarity between movies based on feature vectors.

5.5 Filtering Techniques

- Content-based filtering (based on similarity)
- Genre-based filtering
- Rating-based filtering

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6. Implementation

The system is implemented using Python and the following libraries:

- Pandas for data handling
- Scikit-learn for machine learning
- Streamlit for web interface

The user interacts with the system via a Streamlit app where they can:

- Select a movie
- Choose a genre
- Set minimum rating
- View recommended movies

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7. Results

The system successfully recommends movies that are similar to user preferences. The addition of genre and rating filters improves the relevance and quality of recommendations.

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8. Advantages

- Simple and efficient
- Easy to use interface
- Fast recommendations
- Customizable filters

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9. Limitations

- Does not use user history (no collaborative filtering)
- Depends on dataset quality
- Limited personalization

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10. Future Scope

- Integration with real-time APIs (like TMDB)
- Adding movie posters
- User login system
- Hybrid recommendation system (content + collaborative)

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11. Conclusion

This project demonstrates how machine learning can be used to build a practical recommendation system. It highlights the importance of data preprocessing, feature extraction, and similarity measures in solving real-world problems.